

Pinned group summary: $\mathrm{SL}(n=4)$

Pinning-Sympy automated output

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1 Basic attributes

Group	SL (n=4)
Matrix size	4×4
Rank	3

1.1 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & x_{02} & x_{03} \\ x_{10} & x_{11} & x_{12} & x_{13} \\ x_{20} & x_{21} & x_{22} & x_{23} \\ x_{30} & x_{31} & x_{32} & -x_{00} - x_{11} - x_{22} \end{bmatrix}$$

1.2 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & t_2 & 0 \\ 0 & 0 & 0 & \frac{1}{t_0 t_1 t_2} \end{bmatrix}$$

1.3 Trivial characters

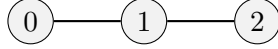
$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A3
Reduced	True
Simply laced	True
Number of roots	12

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
$(-1, 0, 0, 1)$	$(-1, 0, 0, 1)$	No	—	No
$(-1, 0, 1, 0)$	$(-1, 0, 1, 0)$	No	—	No
$(-1, 1, 0, 0)$	$(-1, 1, 0, 0)$	No	—	No
$(0, -1, 0, 1)$	$(0, -1, 0, 1)$	No	—	No
$(0, -1, 1, 0)$	$(0, -1, 1, 0)$	No	—	No
$(0, 0, -1, 1)$	$(0, 0, -1, 1)$	No	—	No
$(0, 0, 1, -1)$	$(0, 0, 1, -1)$	Yes	+	No
$(0, 1, -1, 0)$	$(0, 1, -1, 0)$	Yes	+	No
$(0, 1, 0, -1)$	$(0, 1, 0, -1)$	No	+	No
$(1, -1, 0, 0)$	$(1, -1, 0, 0)$	Yes	+	No
$(1, 0, -1, 0)$	$(1, 0, -1, 0)$	No	+	No
$(1, 0, 0, -1)$	$(1, 0, 0, -1)$	No	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

α	β	Pairs (i, j)
$(-1, 0, 0, 1)$	$(0, 0, 1, -1)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(0, 1, 0, -1)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(1, 0, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(0, 0, -1, 1)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(0, 1, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(1, 0, 0, -1)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(0, -1, 0, 1)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(0, -1, 1, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(1, 0, -1, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 0, 1)$	$(0, 0, 1, -1)$	$(1, 1)$
$(0, -1, 0, 1)$	$(0, 1, -1, 0)$	$(1, 1)$
$(0, -1, 0, 1)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(0, 0, -1, 1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(0, 1, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(1, 0, -1, 0)$	$(1, 1)$
$(0, 0, -1, 1)$	$(0, 1, 0, -1)$	$(1, 1)$
$(0, 0, -1, 1)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, 0, 1, -1)$	$(0, 1, -1, 0)$	$(1, 1)$
$(0, 0, 1, -1)$	$(1, 0, -1, 0)$	$(1, 1)$
$(0, 1, -1, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(0, 1, 0, -1)$	$(1, -1, 0, 0)$	$(1, 1)$